

TASK 3 — DATA LOGGER: SENSOR → SD CARD

1. Aim:

To design and implement a data logging system using an Arduino UNO that measures temperature and humidity using a DHT11 sensor, obtains real-time date and time from a DS3231 RTC module, and stores the sensor readings in CSV format on a microSD card.

2. Goal:

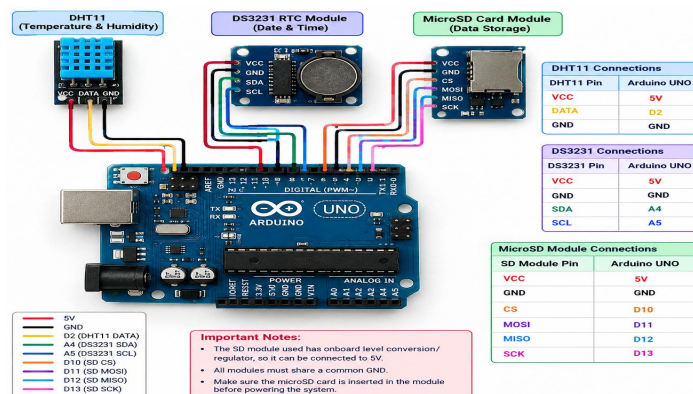
The main goal of this task is to:

- Measure temperature and humidity periodically.
- Obtain accurate date and time using the DS3231 RTC.
- Store the sensor readings with timestamps.
- Save the collected data in a **CSV file** on a microSD card.
- Record at least **100 samples** for analysis.

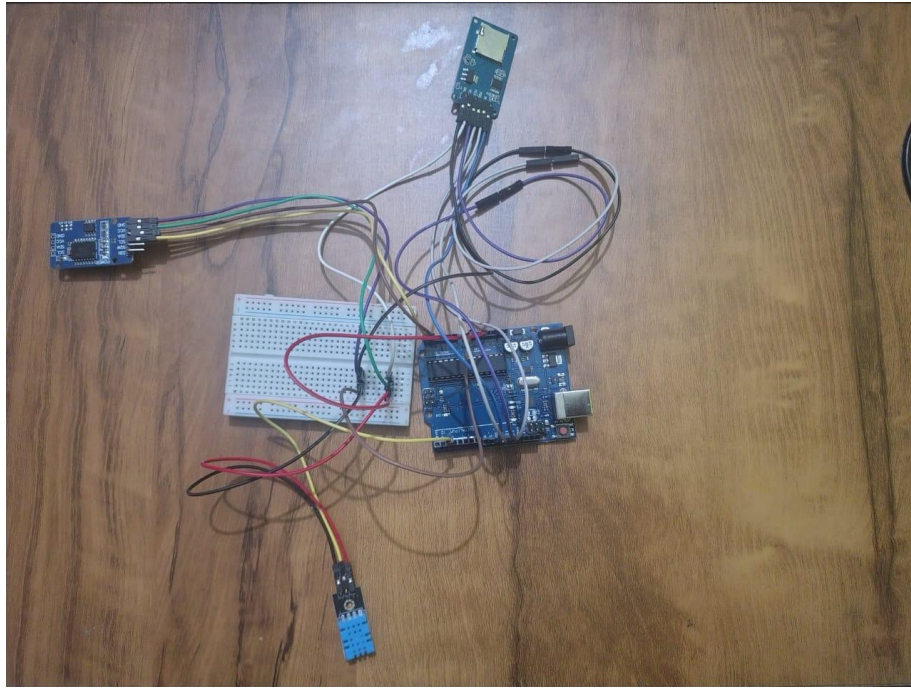
3. Components Required:

S.No.	Component	Quantity
1	Arduino UNO	1
2	DHT11 Temperature & Humidity Sensor	1
3	DS3231 RTC Module	1
4	MicroSD SPI Module with onboard level conversion/regulator	1
5	8 GB MicroSD Card	1
6	Breadboard	1
7	Jumper Wires	As required
8	USB Cable	1

4. Circuit Connections:



5. Complete circuit connection:



6. Program:

```
#include <Wire.h>
#include <RTClib.h>
#include <SPI.h>
#include <SD.h>
#include <DHT.h>

#define DHTPIN 2
#define DHTTYPE DHT11
#define SD_CS 10

DHT dht(DHTPIN, DHTTYPE);
RTC_DS3231 rtc;

int sampleNumber = 0;

void setup() {
  Serial.begin(9600);
  dht.begin();

  if (!rtc.begin()) {
    Serial.println(F("ERROR: DS3231 not found!"));
    while (1);
  }

  if (rtc.lostPower()) {
    Serial.println(F("RTC lost power."));
    rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));
  }
}
```

```

}

Serial.println(F("Initializing SD card..."));

if (!SD.begin(SD_CS)) {
  Serial.println(F("ERROR: SD card initialization failed!"));
  while (1);
}

Serial.println(F("SD card initialized successfully.));

SD.remove("DATA.CSV");

File file = SD.open("DATA.CSV", FILE_WRITE);

if (file) {
  file.println(F("Sample,Date,Time,Temperature_C,Humidity_%"));
  file.close();
  Serial.println(F("DATA.CSV created successfully.));
}
else {
  Serial.println(F("ERROR: Cannot create DATA.CSV!"));
  while (1);
}

Serial.println();
Serial.println(F("=====));
Serial.println(F("  DATA LOGGER STARTED));
Serial.println(F("=====));
Serial.println(F("Interval: 10 seconds));
Serial.println(F("Samples : 100));
Serial.println(F("File   : DATA.CSV));
Serial.println(F("=====));
Serial.println();
}

void loop() {

  if (sampleNumber >= 100) {
    Serial.println();
    Serial.println(F("=====));
    Serial.println(F("  DATA LOGGING COMPLETED));
    Serial.println(F("=====));
    Serial.println(F("100 samples saved.));
    while (1);
  }

  float temperature = dht.readTemperature();
  float humidity = dht.readHumidity();

```

```

if (isnan(temperature) || isnan(humidity)) {
  Serial.println(F("DHT11 reading failed!"));
  delay(2000);
  return;
}

DateTime now = rtc.now();

sampleNumber++;

Serial.print(F("Sample: "));
Serial.print(sampleNumber);

Serial.print(F(" | "));

if (now.day() < 10) Serial.print('0');
Serial.print(now.day());
Serial.print('/');

if (now.month() < 10) Serial.print('0');
Serial.print(now.month());
Serial.print('/');
Serial.print(now.year());

Serial.print(F(" | "));

if (now.hour() < 10) Serial.print('0');
Serial.print(now.hour());
Serial.print(':');

if (now.minute() < 10) Serial.print('0');
Serial.print(now.minute());
Serial.print(':');

if (now.second() < 10) Serial.print('0');
Serial.print(now.second());

Serial.print(F(" | T: "));
Serial.print(temperature);
Serial.print(F(" C | H: "));
Serial.print(humidity);
Serial.println(F(" %"));

File file = SD.open("DATA.CSV", FILE_WRITE);

if (file) {

  file.print(sampleNumber);
  file.print(',');

```

```

file.print(now.day());
file.print('/');
file.print(now.month());
file.print('/');
file.print(now.year());
file.print(',');

if (now.hour() < 10) file.print('0');
file.print(now.hour());
file.print(':');

if (now.minute() < 10) file.print('0');
file.print(now.minute());
file.print(':');

if (now.second() < 10) file.print('0');
file.print(now.second());
file.print(',');

file.print(temperature);
file.print(',');

file.println(humidity);

file.close();

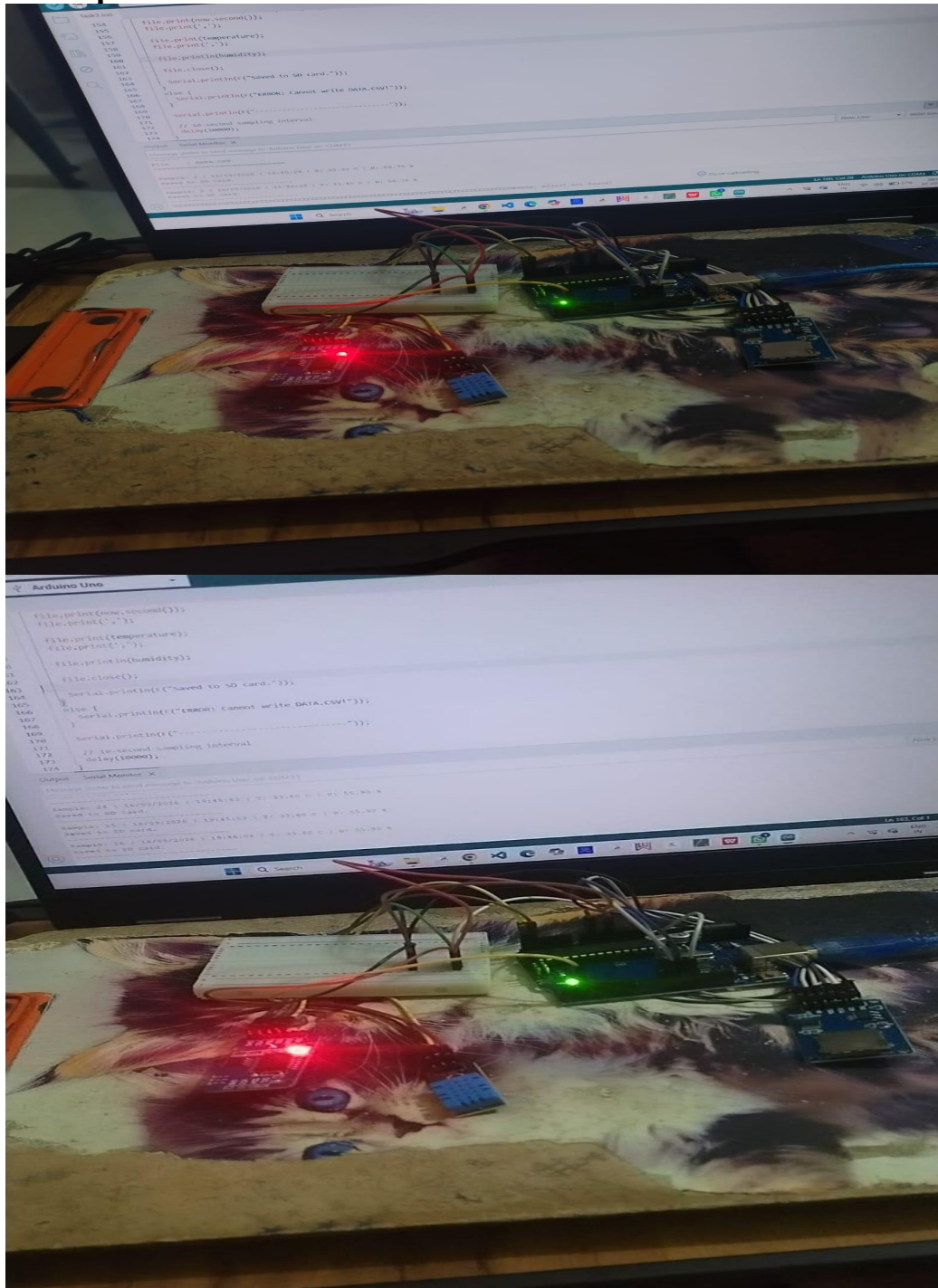
Serial.println(F("Saved to SD card."));
}
else {
  Serial.println(F("ERROR: Cannot write DATA.CSV!"));
}

Serial.println(F("-----"));

delay(10000);
}

```

7. Output:



8. TASK 3 DEMO VIDEO:

<https://drive.google.com/file/d/1x9IJUn6tYfqYoPYruXimaZLZHi1GimLr/view?usp=drivesdk>

9. TEMPERATURE DATA:

[DATA.CSV](#)

10: RESULT:

The Arduino-based data logger was successfully implemented. The DHT11 sensor measured temperature and humidity, while the DS3231 RTC provided date and time information. The Arduino UNO successfully recorded and stored **100 timestamped sensor readings** in DATA.CSV on the microSD card.

Thus, the required sensor-to-SD-card data logging system was successfully completed.